

Task 5: Exploratory Data Analysis (EDA) – Titanic Dataset

Objective

The objective of this analysis is to explore the Titanic dataset using statistical and visual techniques to identify patterns, trends, relationships, and anomalies.

Tools Used

- Python
- Pandas
- Matplotlib
- Seaborn

Dataset

The dataset used for this analysis is the Titanic `train.csv` dataset.

Analysis Performed

The following exploratory techniques are used:

- Dataset structure and information
- Descriptive statistics
- Categorical value counts
- Missing value analysis
- Histograms
- Countplots
- Boxplots
- Scatterplots
- Correlation heatmap
- Pairplot
- Survival rate analysis
- Family-size analysis

1. Import Libraries and Load Dataset

```
In [1]: import pandas as pd

df = pd.read_csv("train.csv")

df.head()
```

Out[1]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500



2. Dataset Overview

```
In [2]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  -
0   PassengerId  891 non-null    int64
1   Survived     891 non-null    int64
2   Pclass       891 non-null    int64
3   Name         891 non-null    object
4   Sex          891 non-null    object
5   Age          714 non-null    float64
6   SibSp        891 non-null    int64
7   Parch        891 non-null    int64
8   Ticket       891 non-null    object
9   Fare         891 non-null    float64
10  Cabin        204 non-null    object
11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

```
In [3]: df.describe()
```

Out[3]:

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

Observation

The Titanic dataset contains 891 passenger records and 12 columns. The dataset includes numerical and categorical variables. Missing values are present mainly in the Age , Cabin , and Embarked columns. The average passenger age is approximately 29.7 years, and approximately 38.4% of passengers survived.

3. Categorical Variable Analysis

3.1 Gender Distribution

```
In [4]: df["Sex"].value_counts()
```

```
Out[4]: Sex
male      577
female    314
Name: count, dtype: int64
```

Observation

The dataset contains 577 male passengers and 314 female passengers. Therefore, male passengers make up the larger proportion of the dataset.

3.2 Survival Distribution

```
In [5]: df["Survived"].value_counts()
```

```
Out[5]: Survived
0       549
1       342
Name: count, dtype: int64
```

Observation

Out of 891 passengers, 549 did not survive and 342 survived. Therefore, approximately 38.4% of passengers survived, while 61.6% did not survive.

3.3 Passenger Class Distribution

```
In [6]: df["Pclass"].value_counts()
```

```
Out[6]: Pclass
3       491
1       216
2       184
Name: count, dtype: int64
```

Observation

The majority of passengers belonged to 3rd class, with 491 passengers. There were 216 passengers in 1st class and 184 passengers in 2nd class.

4. Missing Value Analysis

```
In [7]: df.isnull().sum()
```

```
Out[7]: PassengerId      0
        Survived        0
        Pclass          0
        Name            0
        Sex             0
        Age            177
        SibSp           0
        Parch           0
        Ticket          0
        Fare            0
        Cabin          687
        Embarked        2
        dtype: int64
```

```
In [8]: df.dtypes
```

```
Out[8]: PassengerId      int64
        Survived        int64
        Pclass          int64
        Name            object
        Sex             object
        Age            float64
        SibSp          int64
        Parch          int64
        Ticket          object
        Fare          float64
        Cabin          object
        Embarked        object
        dtype: object
```

Observation

Missing values are present in three columns. The `Cabin` column has the highest number of missing values with 687, followed by `Age` with 177 and `Embarked` with 2. The `Cabin` column has a substantial amount of missing data and may require special consideration in further analysis.

5. Duplicate Value Analysis

```
In [28]: df.duplicated().sum()
```

```
Out[28]: np.int64(0)
```

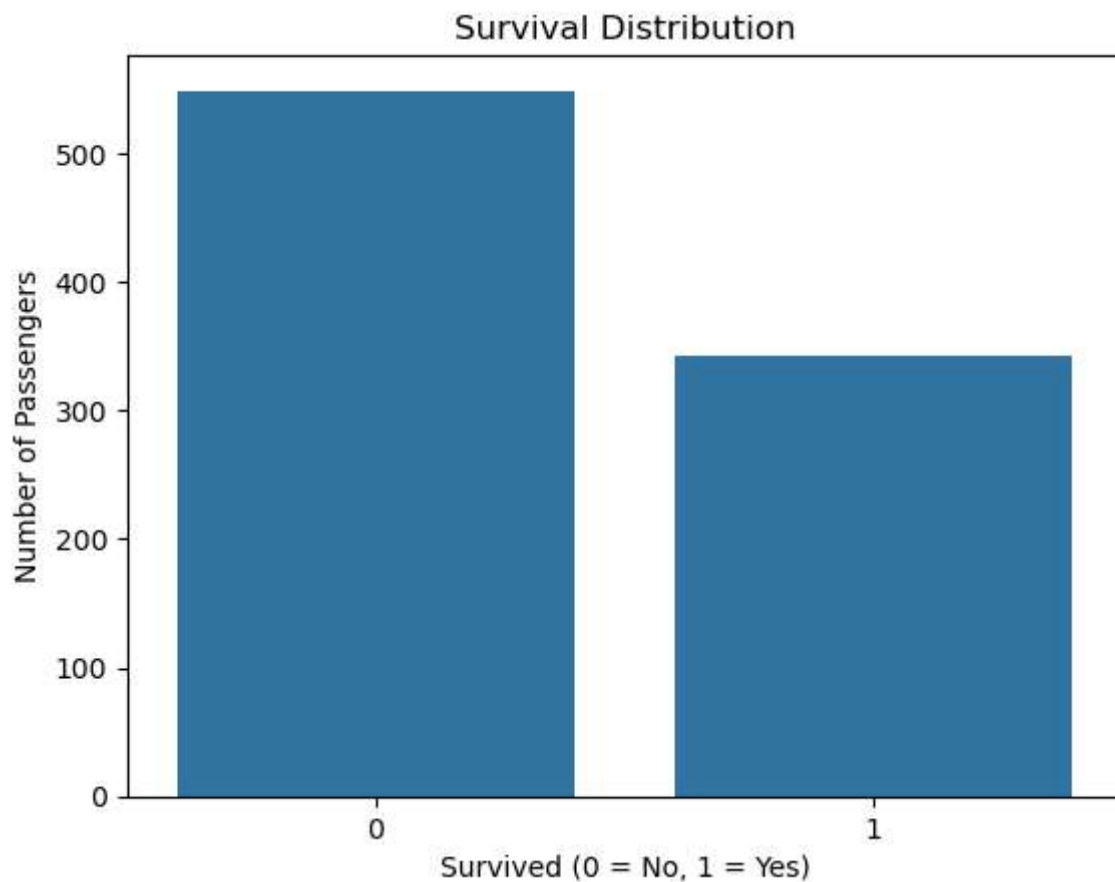
Observation The dataset contains no duplicate rows.

6.1 Survival Distribution

```
In [11]: sns.countplot(data=df, x="Survived")

plt.title("Survival Distribution")
plt.xlabel("Survived (0 = No, 1 = Yes)")
```

```
plt.ylabel("Number of Passengers")  
plt.show()
```

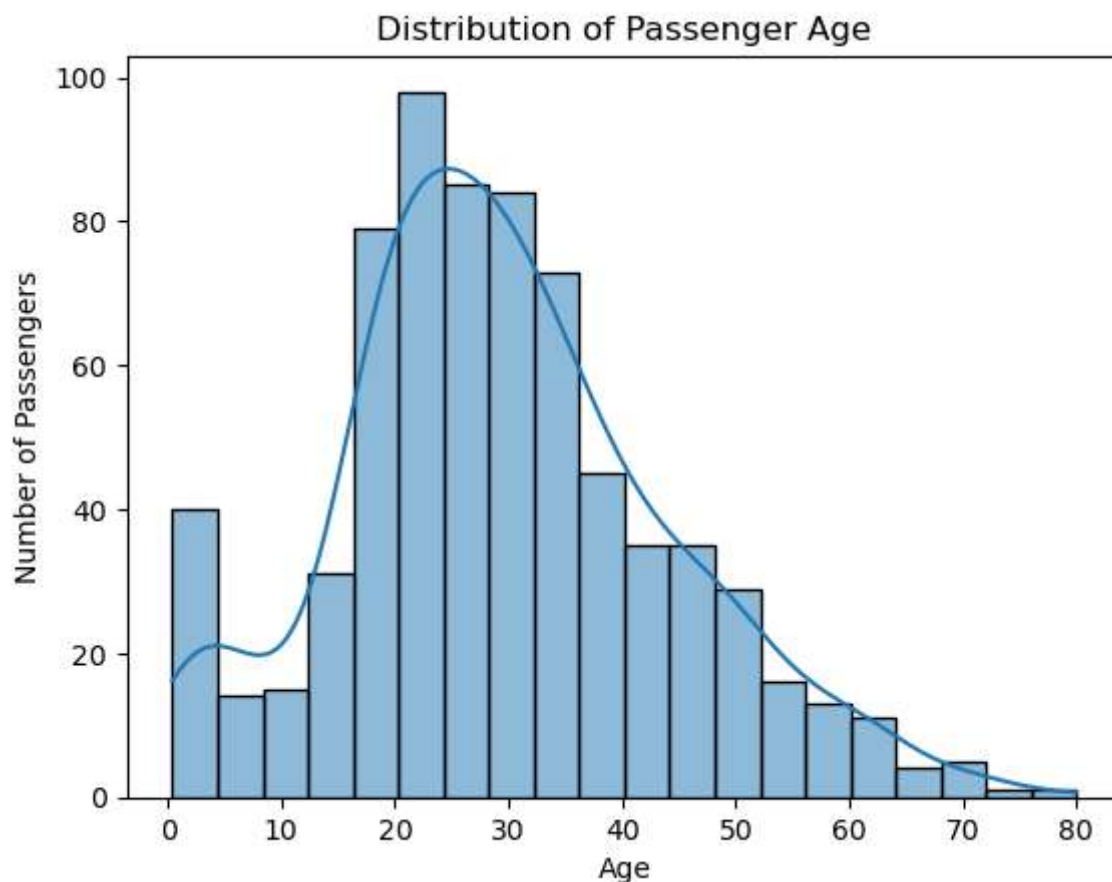


Observation

The graph shows that 549 passengers did not survive, while 342 passengers survived. Therefore, the majority of passengers did not survive. Approximately 38.4% of passengers survived, while 61.6% did not survive.

6.2 Age Distribution

```
In [10]: import matplotlib.pyplot as plt  
import seaborn as sns  
  
sns.histplot(data=df, x="Age", bins=20, kde=True)  
  
plt.title("Distribution of Passenger Age")  
plt.xlabel("Age")  
plt.ylabel("Number of Passengers")  
  
plt.show()
```



Observation

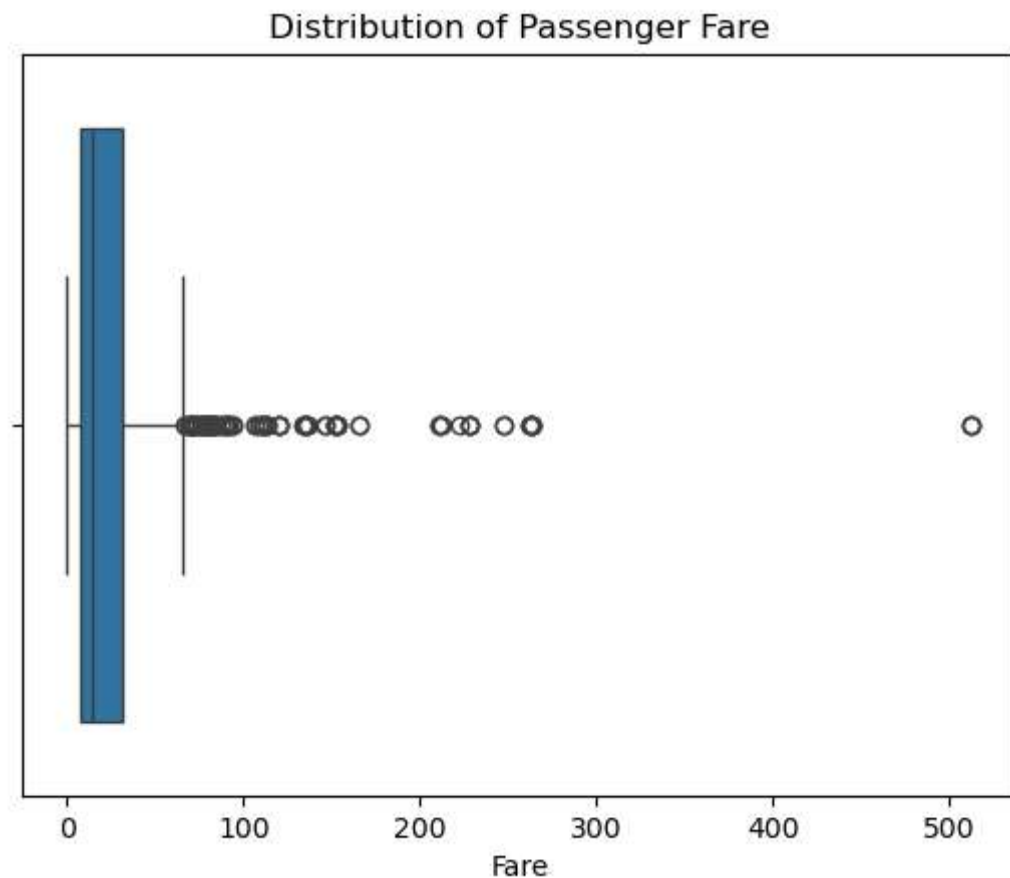
The age distribution is concentrated mainly between approximately 20 and 40 years. There are fewer very young and older passengers. The distribution also contains missing values because 177 passengers do not have recorded ages.

6.3 Fare Distribution

```
In [15]: sns.boxplot(data=df, x="Fare")

plt.title("Distribution of Passenger Fare")
plt.xlabel("Fare")

plt.show()
```



Observation

The fare distribution is highly right-skewed. Most passengers paid relatively low fares, while a small number of passengers paid very high fares. Several potential outliers are visible in the boxplot. The maximum fare was approximately 512.33, while the median fare was approximately 14.45.

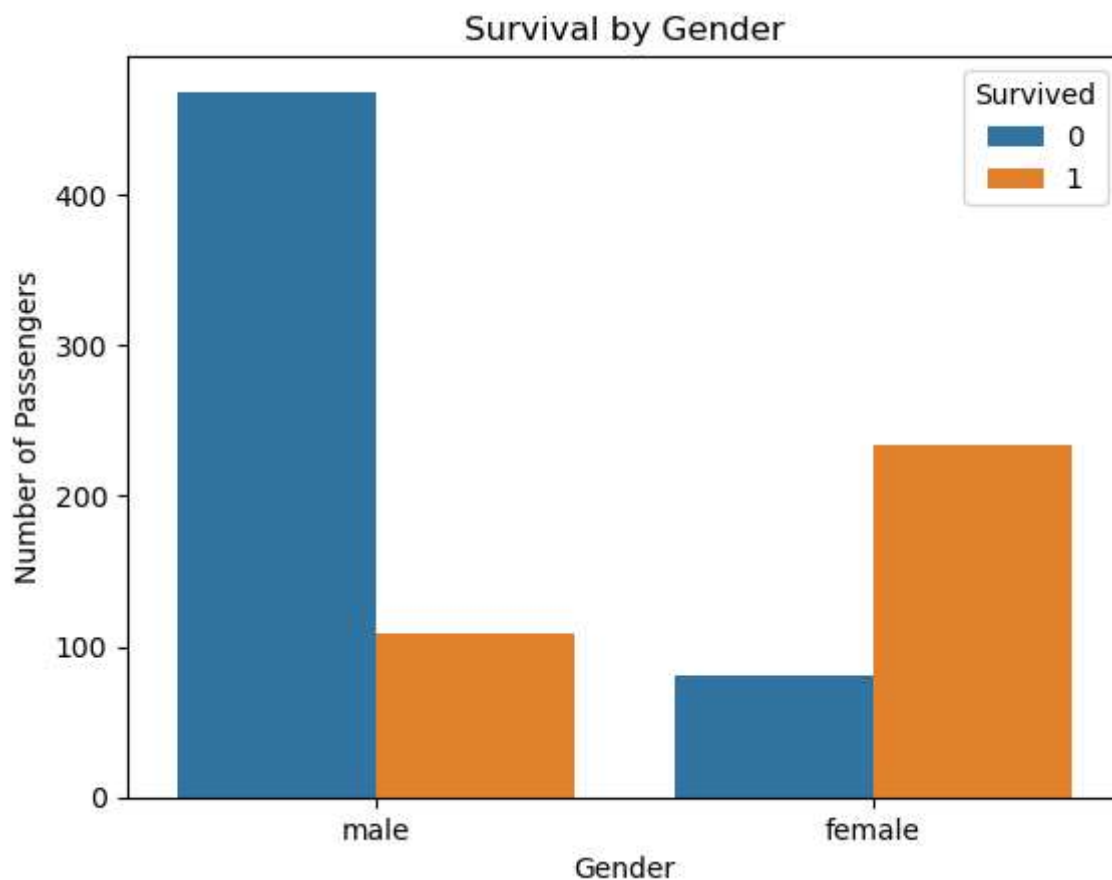
7. Bivariate Analysis

7.1 Survival by Gender

```
In [12]: sns.countplot(data=df, x="Sex", hue="Survived")

plt.title("Survival by Gender")
plt.xlabel("Gender")
plt.ylabel("Number of Passengers")

plt.show()
```

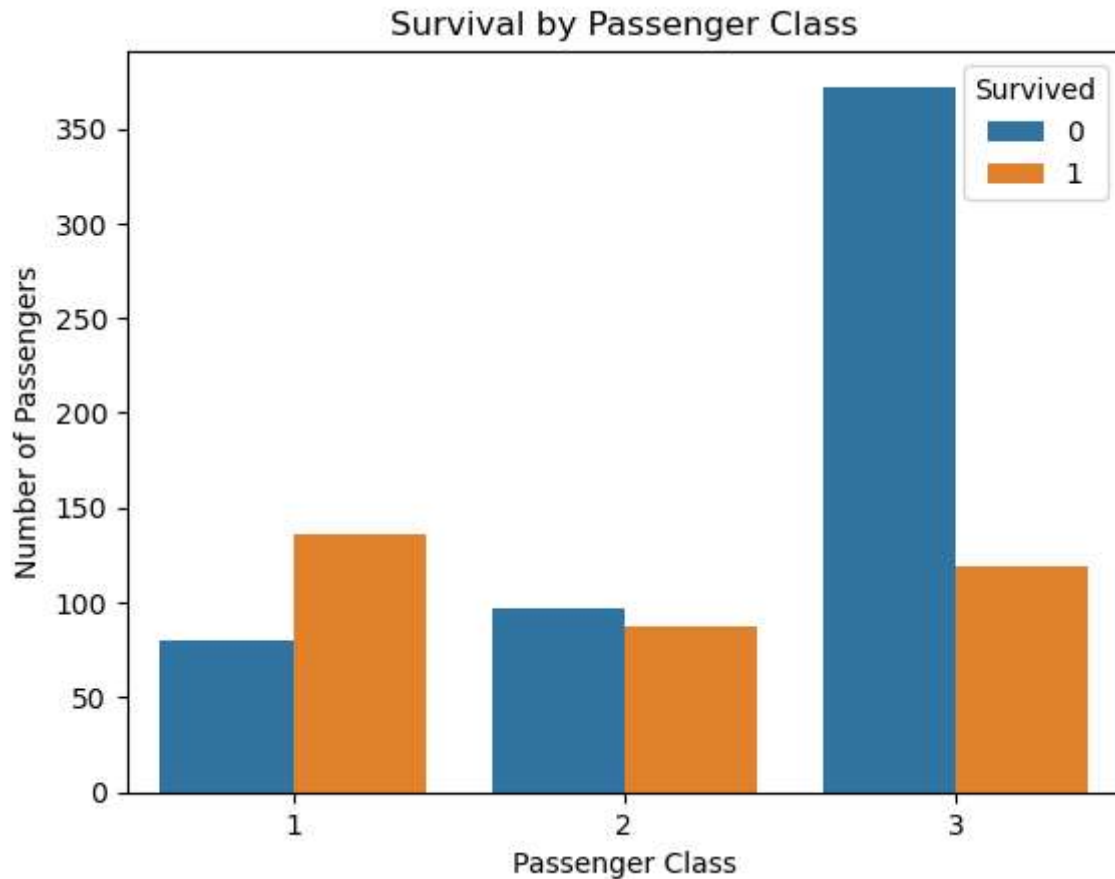


7.2 Survival by Passenger Class

```
In [13]: sns.countplot(data=df, x="Pclass", hue="Survived")

plt.title("Survival by Passenger Class")
plt.xlabel("Passenger Class")
plt.ylabel("Number of Passengers")

plt.show()
```



Observation

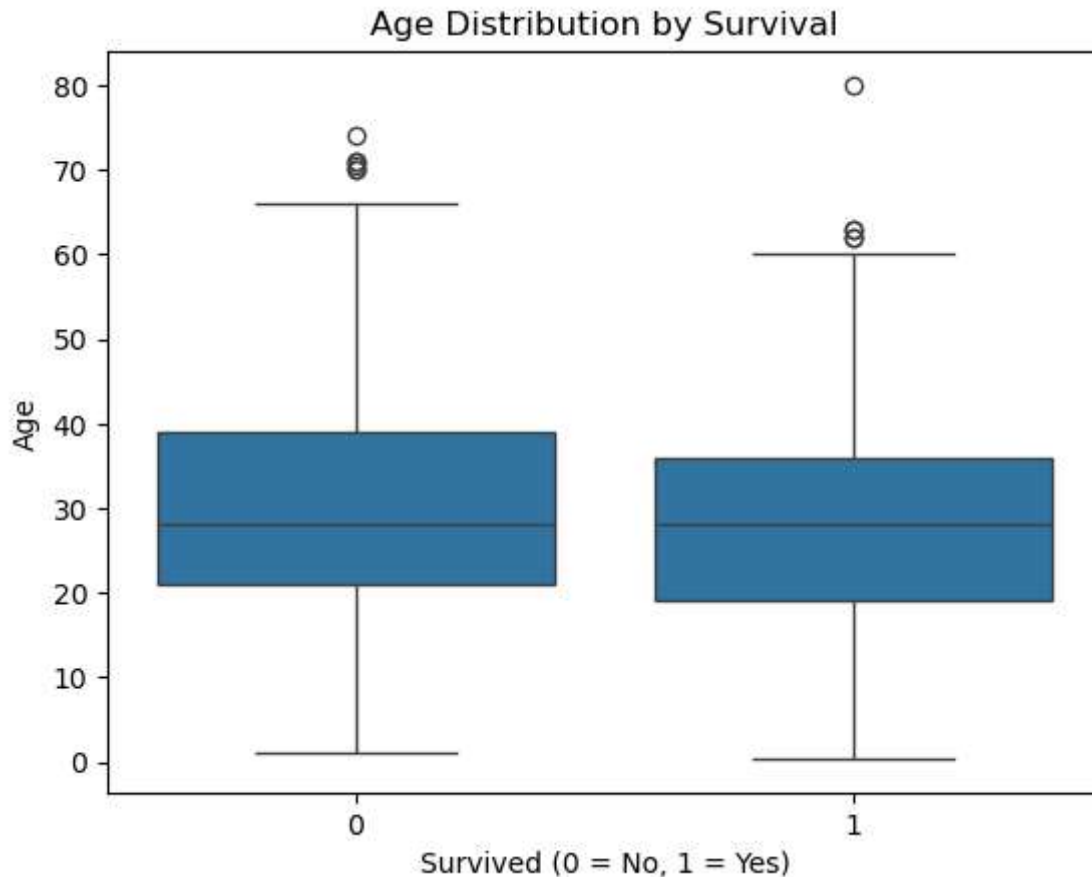
Survival varied considerably across passenger classes. First-class passengers had the highest survival rate at approximately 63.0%, followed by second-class passengers at approximately 47.3%. Third-class passengers had the lowest survival rate at approximately 24.2%. This indicates a strong association between passenger class and survival.

7.3 Age Distribution by Survival

```
In [14]: sns.boxplot(data=df, x="Survived", y="Age")

plt.title("Age Distribution by Survival")
plt.xlabel("Survived (0 = No, 1 = Yes)")
plt.ylabel("Age")

plt.show()
```



Observation

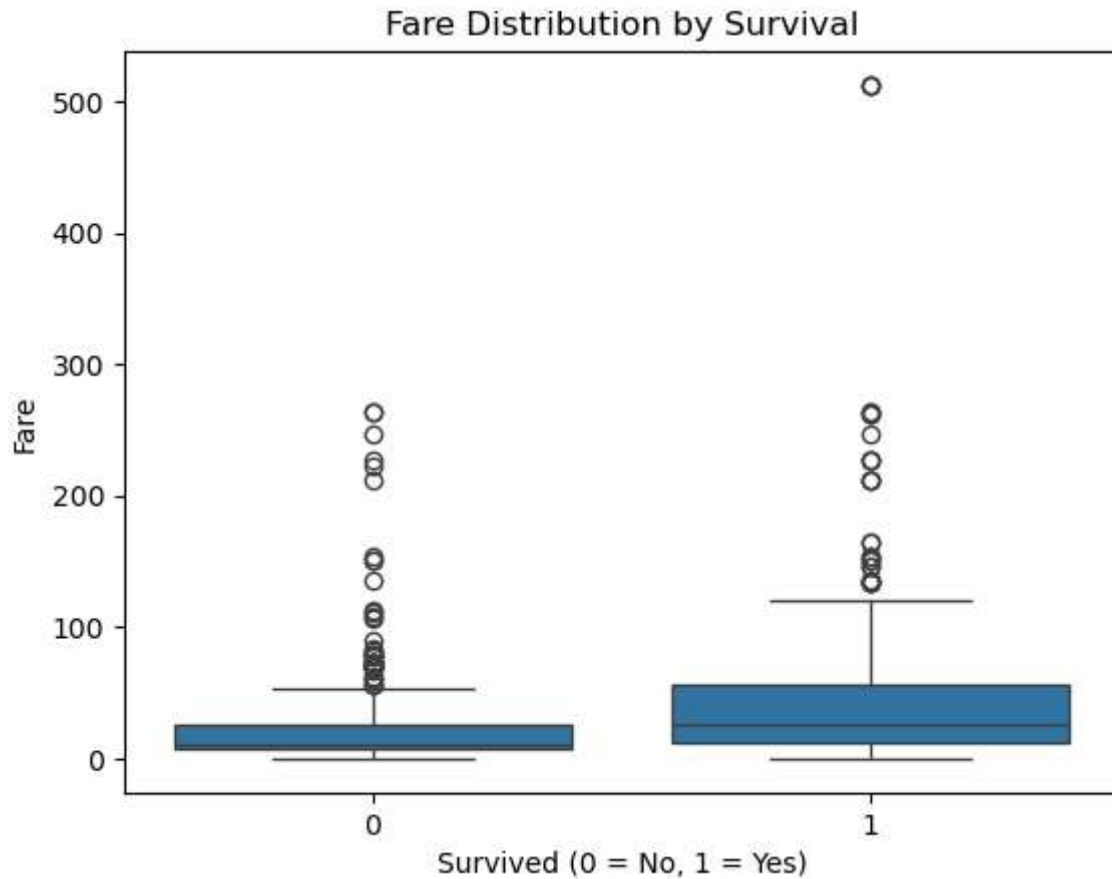
The age distributions of survivors and non-survivors overlap considerably. This suggests that age alone does not clearly distinguish between passengers who survived and those who did not. However, there are some differences in the median ages and overall distributions of the two groups. The analysis should also consider the missing values in the Age column.

7.4 Fare Distribution by Survival

```
In [16]: sns.boxplot(data=df, x="Survived", y="Fare")

plt.title("Fare Distribution by Survival")
plt.xlabel("Survived (0 = No, 1 = Yes)")
plt.ylabel("Fare")

plt.show()
```



Observation

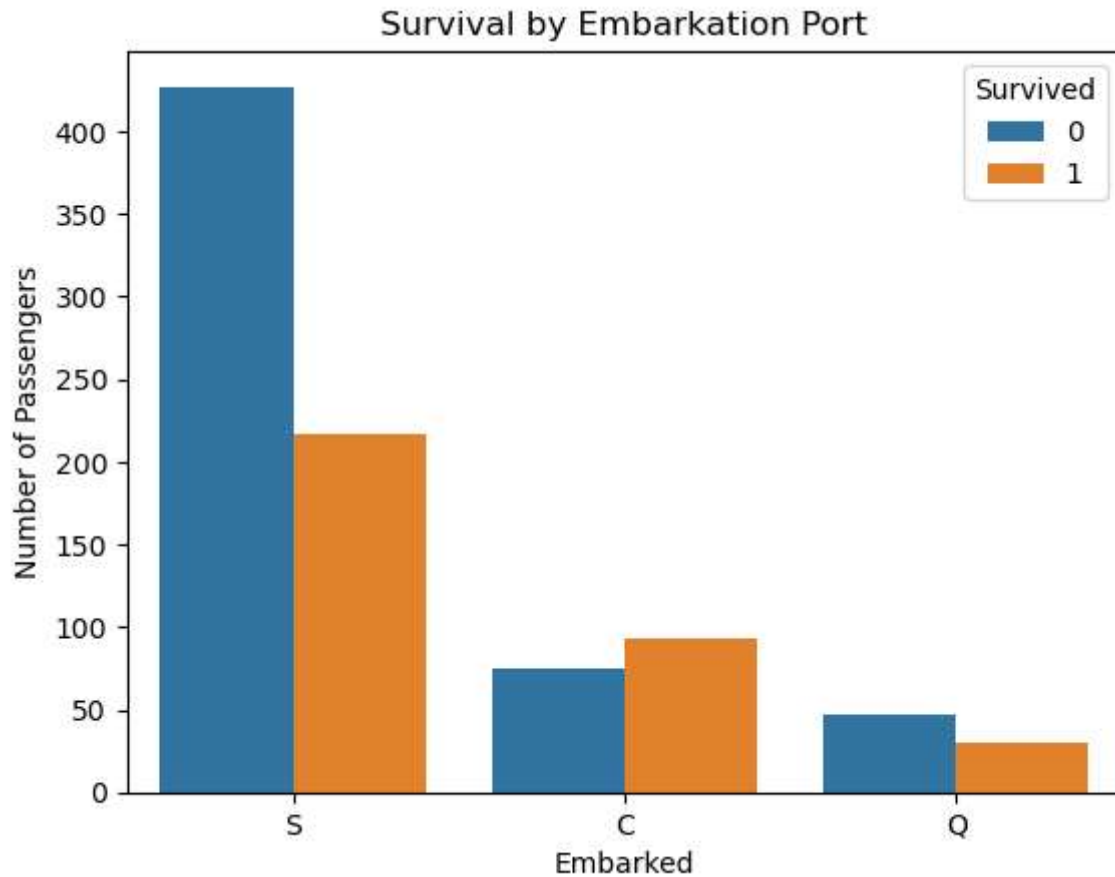
Passengers who survived generally paid higher fares than passengers who did not survive. The median fare for survivors is higher, suggesting a relationship between fare and survival. The plot also shows several high-fare outliers.

7.5 Survival by Embarkation Port

```
In [26]: sns.countplot(data=df, x="Embarked", hue="Survived")

plt.title("Survival by Embarkation Port")
plt.xlabel("Embarked")
plt.ylabel("Number of Passengers")

plt.show()
```



Observation

Survival varied across the three embarkation ports. Passengers who embarked from port C had the highest survival rate at approximately 55.4%, followed by port Q at approximately 39.0% and port S at approximately 33.7%. This suggests an association between embarkation port and survival, although the difference may also be influenced by other factors such as passenger class and gender.

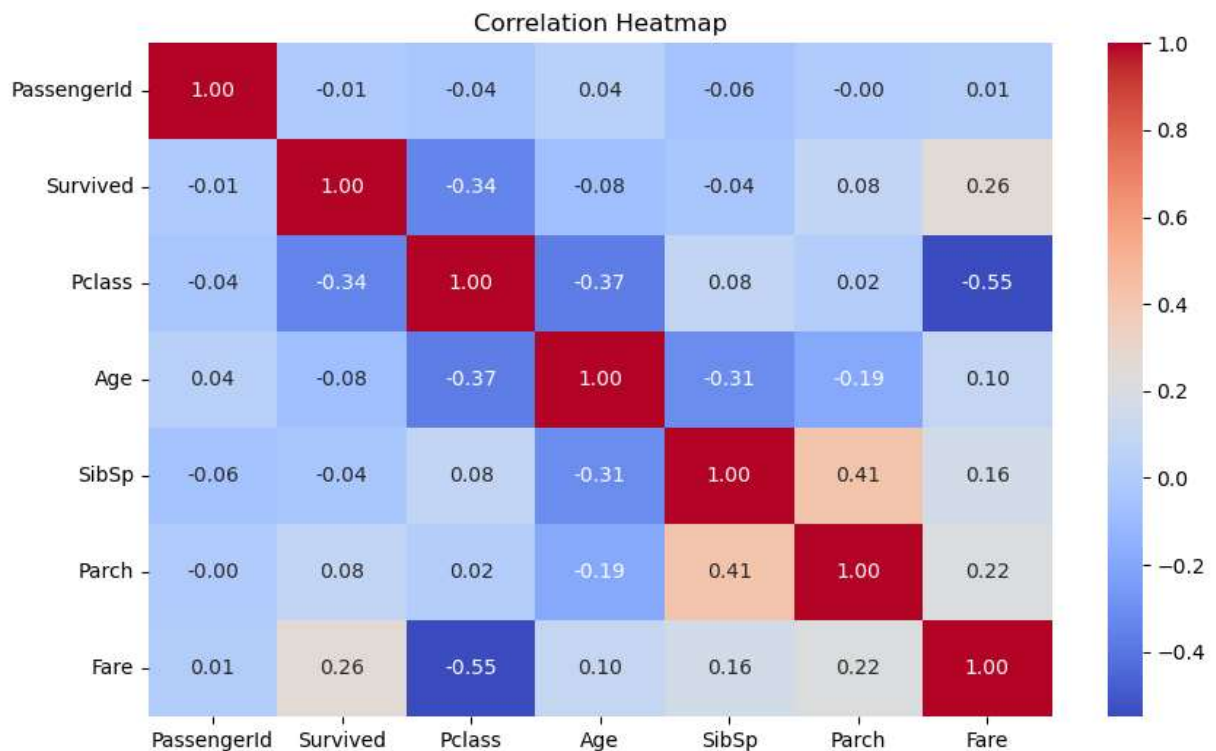
8. Multivariate Analysis

8.1 Correlation Heatmap

```
In [17]: plt.figure(figsize=(10, 6))

sns.heatmap(
    df.corr(numeric_only=True),
    annot=True,
    cmap="coolwarm",
    fmt=".2f"
)

plt.title("Correlation Heatmap")
plt.show()
```



Observation

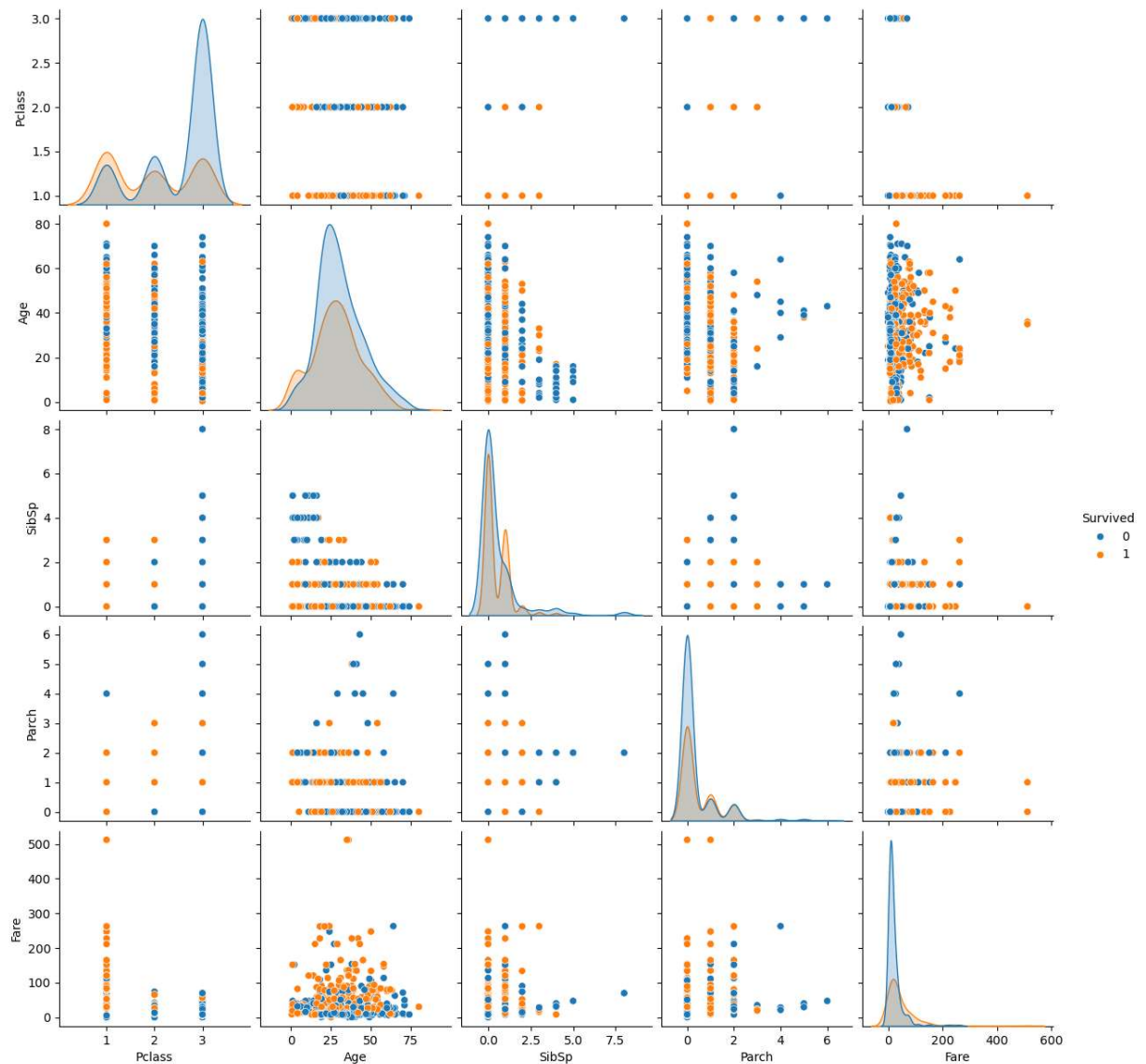
The heatmap shows the correlation between the numerical variables in the Titanic dataset.

`Pclass` has a negative correlation with `Survived`, indicating that higher class numbers (representing lower passenger classes) are associated with lower survival. `Fare` has a positive correlation with `Survived`, suggesting that passengers who paid higher fares generally had better survival outcomes. Most other numerical variables show relatively weak correlations with survival.

8.2 Pairplot

```
In [19]: sns.pairplot(
          df[["Survived", "Pclass", "Age", "SibSp", "Parch", "Fare"]],
          hue="Survived"
        )

plt.show()
```



Observation

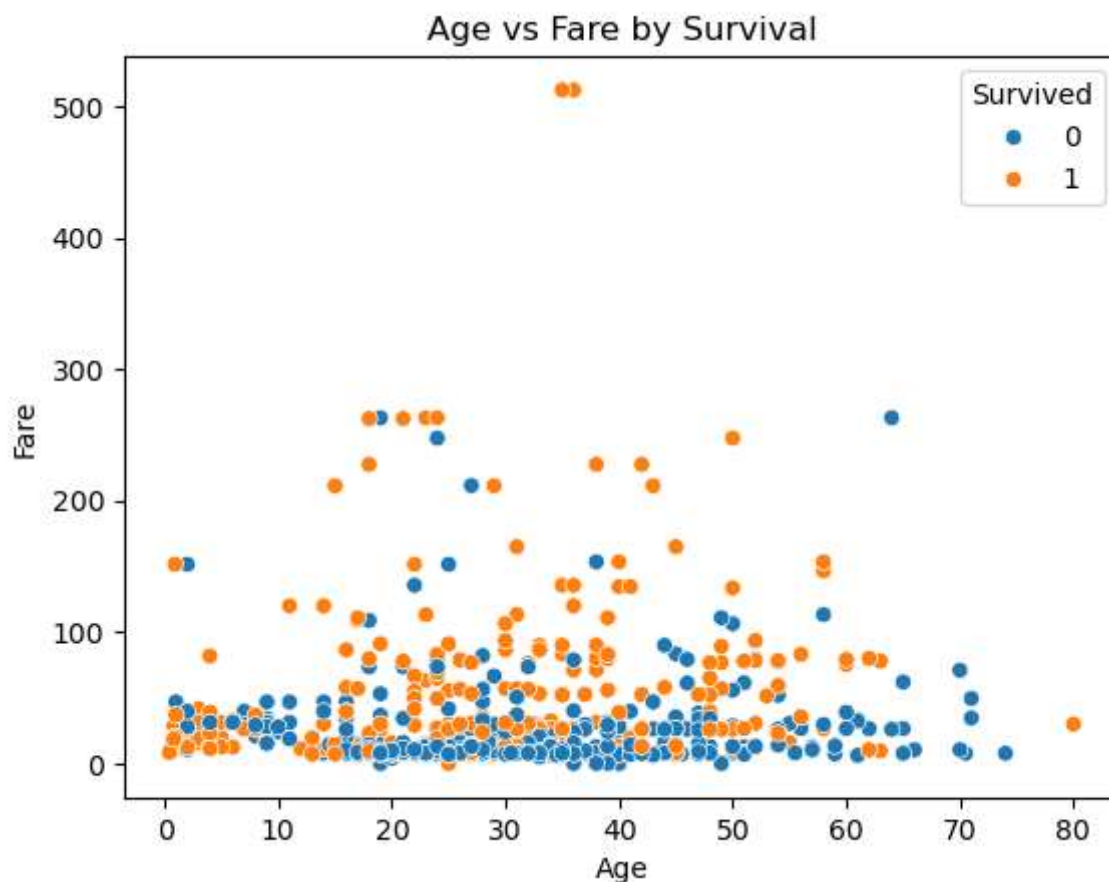
The pairplot shows relationships among the numerical variables and helps identify patterns and distributions for survivors and non-survivors. `Pclass` and `Fare` show noticeable relationships with survival, while `Age` shows considerable overlap between the two survival groups. The plot also highlights the presence of extreme fare values.

8.3 Age vs Fare by Survival

```
In [18]: sns.scatterplot(data=df, x="Age", y="Fare", hue="Survived")

plt.title("Age vs Fare by Survival")
plt.xlabel("Age")
plt.ylabel("Fare")

plt.show()
```



Observation

The scatterplot shows the relationship between passenger age and fare, with different colors representing survival status. There is no clear linear relationship between age and fare. However, passengers who paid higher fares appear in both survival groups, and some high-fare observations are visible as outliers.

Survival Rate by Gender

```
In [20]: df.groupby("Sex")["Survived"].mean() * 100
```

```
Out[20]: Sex
female    74.203822
male      18.890815
Name: Survived, dtype: float64
```

Survival Rate by Passenger Class

```
In [21]: df.groupby("Pclass")["Survived"].mean() * 100
```

```
Out[21]: Pclass
1      62.962963
2      47.282609
3      24.236253
Name: Survived, dtype: float64
```

9.1 Creating Family Size

```
In [22]: df["FamilySize"] = df["SibSp"] + df["Parch"] + 1

df[["SibSp", "Parch", "FamilySize"]].head()
```

```
Out[22]:
```

	SibSp	Parch	FamilySize
0	1	0	2
1	1	0	2
2	0	0	1
3	1	0	2
4	0	0	1

9.2 Survival Rate by Family Size

```
In [23]: df.groupby("FamilySize")["Survived"].mean() * 100
```

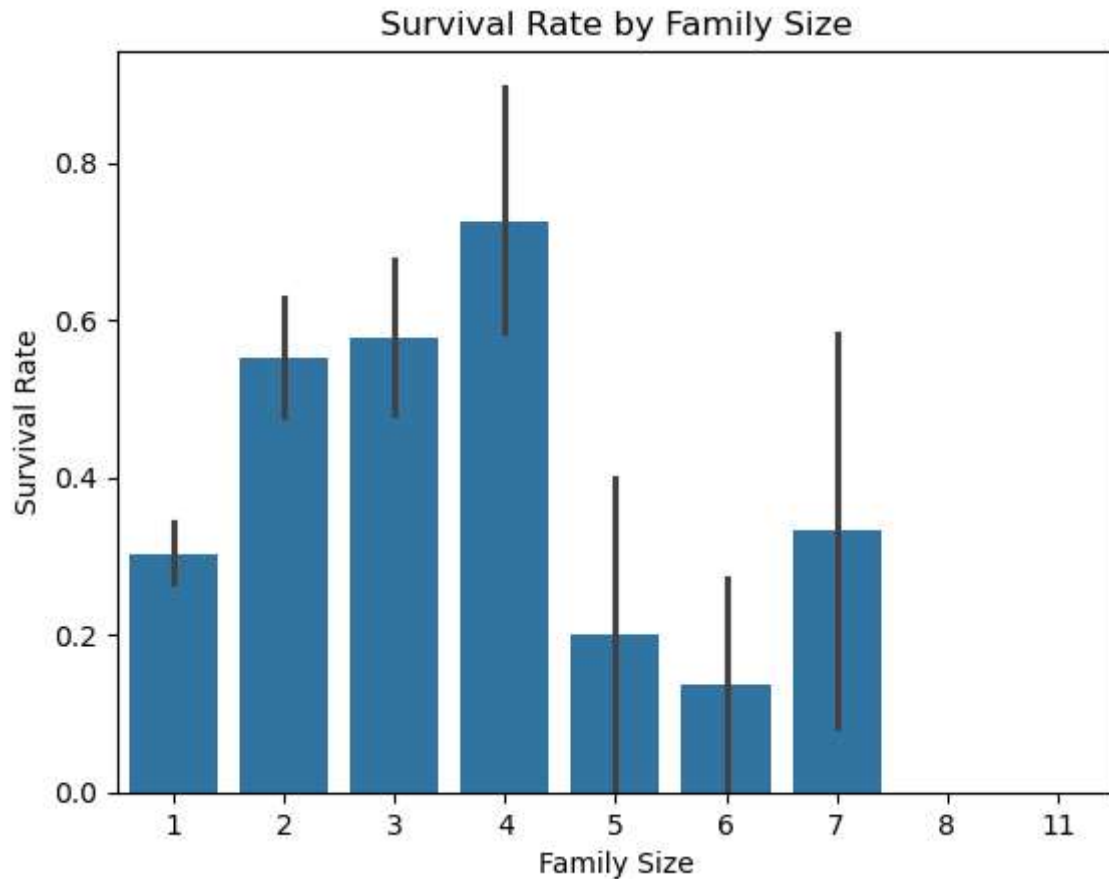
```
Out[23]: FamilySize
1      30.353818
2      55.279503
3      57.843137
4      72.413793
5      20.000000
6      13.636364
7      33.333333
8       0.000000
11     0.000000
Name: Survived, dtype: float64
```

Survival Rate by Family Size — Visualization

```
In [24]: sns.barplot(
    data=df,
    x="FamilySize",
    y="Survived"
)

plt.title("Survival Rate by Family Size")
plt.xlabel("Family Size")
plt.ylabel("Survival Rate")

plt.show()
```



```
In [25]: df["Embarked"].value_counts()
```

```
Out[25]: Embarked
S      644
C      168
Q       77
Name: count, dtype: int64
```

Survival Rate by Embarkation Port

```
In [27]: df.groupby("Embarked")["Survived"].mean() * 100
```

```
Out[27]: Embarked
C      55.357143
Q      38.961039
S      33.695652
Name: Survived, dtype: float64
```

Survival Rate by Gender and Passenger Class

```
In [29]: pd.crosstab(
    [df["Pclass"], df["Sex"]],
    df["Survived"],
    normalize="index"
) * 100
```

Out[29]:

		Survived	0	1
Pclass	Sex			
1	female	3.191489	96.808511	
	male	63.114754	36.885246	
2	female	7.894737	92.105263	
	male	84.259259	15.740741	
3	female	50.000000	50.000000	
	male	86.455331	13.544669	

10. Key Findings and Conclusion

Key Findings

- The dataset contains 891 passengers and 12 columns.
- There are 177 missing values in `Age` , 687 missing values in `Cabin` , and 2 missing values in `Embarked` .
- 342 passengers survived, while 549 passengers did not survive.
- Female passengers had a much higher survival rate (approximately 74.2%) than male passengers (approximately 18.9%).
- First-class passengers had the highest survival rate (approximately 63.0%), while third-class passengers had the lowest (approximately 24.2%).
- Passengers who paid higher fares generally had better survival outcomes.
- Passengers embarking from port C had the highest survival rate among the three ports.
- Passengers traveling in small family groups generally had better survival rates than those traveling alone or in very large families.
- The analysis of gender and passenger class showed that both factors were strongly associated with survival.
- The Fare variable contains several high-value outliers, while the Age distribution contains missing values.

Conclusion

The exploratory analysis shows that survival on the Titanic was strongly associated with factors such as gender, passenger class, and fare. Female passengers and passengers in higher classes generally had better chances of survival. Family size and embarkation port also showed differences in survival rates. Overall, the EDA helped identify important patterns, relationships, missing values, and outliers in the Titanic dataset.

In []: